

Additional Experimental Result

March 2026

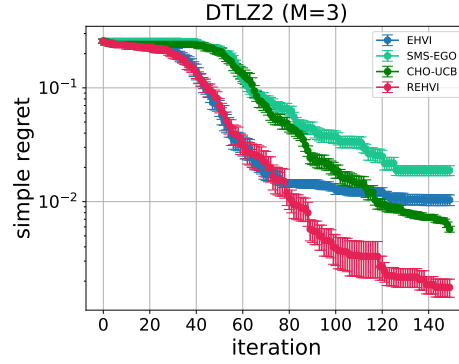


Figure 1: DTLZ2 with 6D input and 4 objectives: REHVI and CHO-UCB perform well relative to their base methods.

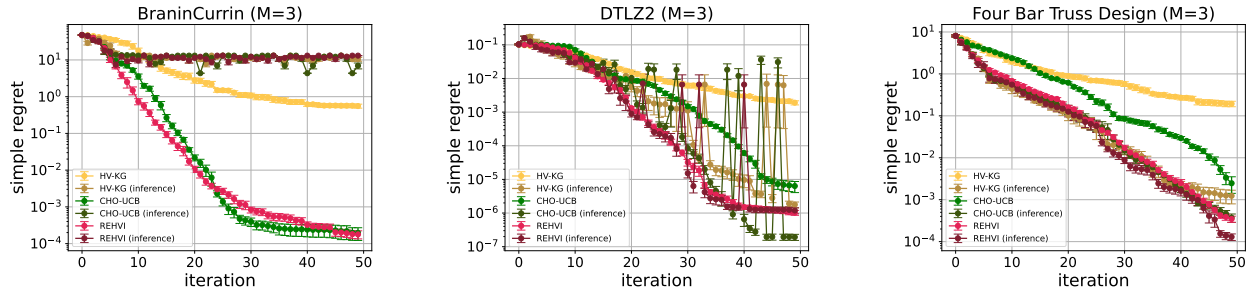


Figure 2: HVKG does not outperform the proposed methods in terms of observation regret. HVKG obtains significantly worse solutions at the same evaluation cost than the proposed methods CHO-UCB and REHVI. In terms of inference regret, all three acquisition functions perform similarly.

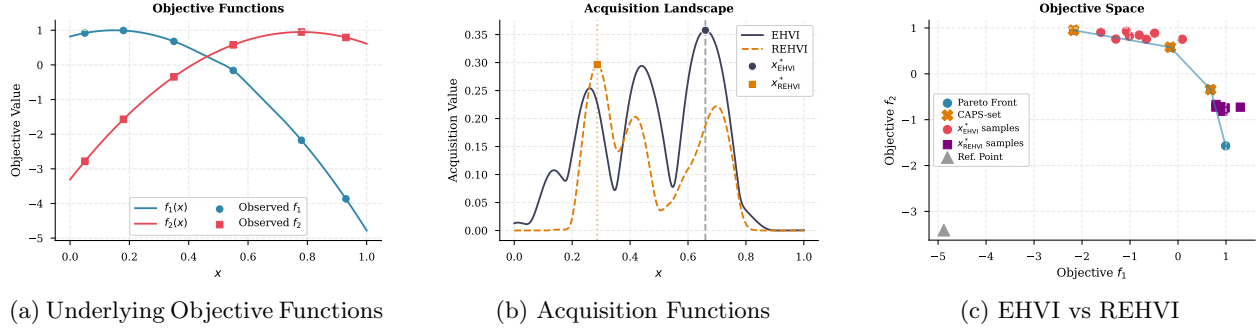


Figure 3: Illustration of EHVI and REHVI acquisition functions on a bi-objective toy problem: objective landscapes, acquisition values, and resulting candidate solutions in objective space.

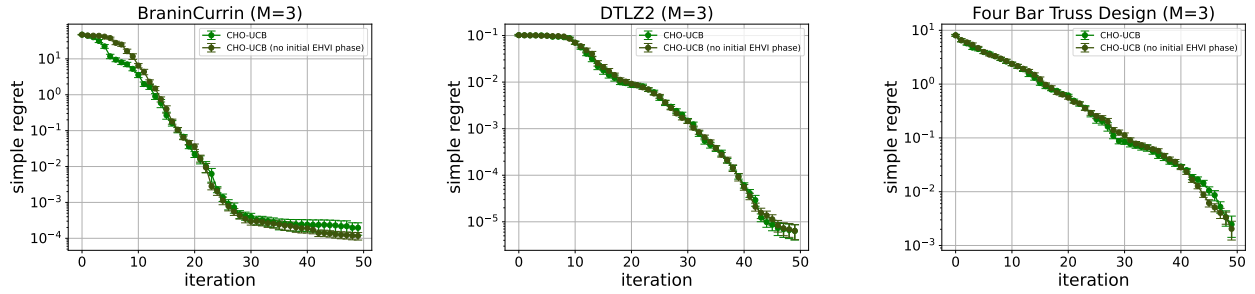


Figure 4: We evaluate CHO-UCB with and without the initial EHVI phase for three benchmarks and observe that it does not impact the convergence behavior. And it makes a difference for the initial performance for only one of the three benchmarks.